OFFICIAL TECHNICAL SPECIFICATION DATASHEET
MANUFACTURER: WAAREE ENERGIES
MODEL / SERIES: Bi-55-540 (Ahnay Series)
DOCUMENT TITLE: Waaree Ahnay Series Bi-55 Framed Dual Glass Bifacial Module Datasheet
REVISION / DATE: DS-BI-55-Rev03-2022 | Published: 2022-09-01

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1. PRODUCT IDENTIFICATION & ARCHITECTURE

Cell Technology:	Monocrystalline PERC
Module Construction:	Bifacial (Glass-Glass)
Number of Cells / Format:	144 Half-cut Mono PERC Cells (M10 91x182 mm format)
Glass / Front Encapsulant:	2.0 mm Low Iron HTAR semi-tempered front & back glass
Frame Material:	Anodized Aluminium Alloy
Junction Box / Protection:	IP68 weatherproof with 3 bypass diodes

2. ELECTRICAL CHARACTERISTICS AT STC (1000 W/m2, 25C, AM 1.5)

Nominal Maximum Power (Pmax / Pnom):	540.0 Wp
Module Aperture Efficiency:	20.98 %
Open Circuit Voltage (Voc):	49.61 V
Short Circuit Current (Isc):	13.83 A
Maximum Power Voltage (Vmp):	41.75 V
Maximum Power Current (Imp):	12.95 A
Maximum System Voltage:	1500.0 V DC
Maximum Series Fuse Rating:	30.0 A

3. TEMPERATURE CHARACTERISTICS & OPERATING LIMITS

Temp. Coefficient of Pmax (gamma):	-0.340 %/C
Temp. Coefficient of Voc (beta):	-0.250 %/C
Temp. Coefficient of Isc (alpha):	+0.050 %/C
Nominal Cell Temperature (NOCT/NMOT):	43 +/- 2 C (NOCT)
Operating Temperature Range:	-40 C to +85 C

4. MECHANICAL DIMENSIONS & RATINGS

Length:	2272.0 mm
Width:	1133.0 mm
Thickness (Height):	35.0 mm
Gross Aperture Area:	2.5742 m2
Module Weight:	32.5 kg
Mechanical Design Load (Front):	5400 Pa (Snow load)
Mechanical Design Load (Rear):	2400 Pa (Wind load)

5. WARRANTY & LIFETIME PERFORMANCE

Product Workmanship Warranty:	12.0 Years
Linear Power Output Warranty:	30.0 Years
Annual Linear Degradation:	0.45 % / year
End of Warranty Retained Output:	84.8 %

6. PROVENANCE & ARCHIVAL METADATA

Authoritative URL:	https://www.waaree.com/datasheets/Waaree_Bi-55_Dual_Glass_Bifacial.pdf
Document Verification:	OFFICIAL_FAMILY_DOCUMENT
Installation Mapping:	BENCHMARK_REFERENCE_MODULE
Verification Date:	2026-11-10

SolarSelect MLOps Hardware Database Reference Standard IEC 61215 / IEC 61730